

The gpt5_6 16-Component Spinor Cosmology

Action, stress tensor, CPL reconstruction, numerical solution, and tests

Reproducible computational report

17 September 2026

Abstract

We define gpt5_6 as a four-flavor Dirac multiplet in $\mathbb{C}^{16}$ with a nonlinear potential depending on the scalar bilinear $S = \bar{\Psi}\Psi$. In a homogeneous flat FLRW spacetime, the Dirac equations give $S = S_0 a^{-3}$ and a perfect-fluid stress tensor with $\rho_\Psi = U$, $p_\Psi = SU_{,S} - U$, and $w_\Psi = -1 + SU_{,S}/U$. We reconstruct a potential that realizes the central CPL history $(w_0, w_a) = (-0.861, -0.60)$ reported for the combined Unite+BAO+CMB analysis by Camilleri et al. A DOP853 integration verifies the bilinear, density, and Friedmann identities at relative errors below 4.6×10^{-11} . The central reconstructed history crosses $w = -1$ and the homogeneous null-energy boundary at $a \simeq 0.76833$ and begins accelerating at $z \simeq 0.74842$. We distinguish this result from the separate Unite-only constant- $w = -0.764$ fit and state the limits of the background reconstruction.

Contents

1	Scope and claim boundary	2
2	Observational inputs	2
3	The 16-component field	2
4	Action, field equation, and stress tensor	3
5	Kinetic density and thermodynamics	3
6	Exact CPL potential reconstruction	4
7	Numerical initial-value problem	4
7.1	Variables and equations	4
7.2	Discriminating numerical checks	5
8	Numerical results	5
8.1	Phantom crossing and the null-energy boundary	5
8.2	Acceleration transitions	6
9	Dark energy and dark matter	9
9.1	Dark-energy interpretation	9
9.2	Dark-matter limit	9
10	Limitations and open work	9

1 Scope and claim boundary

This report supplies a field-theoretic mechanism for a time-varying dark-energy background and a complete reproducible numerical calculation. It does not perform an observational likelihood analysis, propagate parameter covariances, introduce a new irreducible Lorentz representation, or prove the stability of perturbations. The potential is reconstructed from a prescribed background history. Consequently, success means that the field equations and stress tensor consistently realize that history, not that nature selects this specific microscopic model.

2 Observational inputs

The supplied source was a six-page Gmail print discussing the results of Camilleri et al. [1]. Because it is not the complete paper, the parameter values and fit labels were checked against the official arXiv v2 record. The two relevant fits are distinct (Table 1).

Table 1: Published constraints used or shown in this project. Errors are the reported asymmetric 68.27% credible intervals.

Data	Model	Ω_m	Dark-energy parameters
Unite only	flat w CDM	$0.197^{+0.056}_{-0.054}$	$w = -0.764^{+0.078}_{-0.096}$
Unite+BAO+CMB	flat $w_0 w_a$ CDM	0.305 ± 0.004	$w_0 = -0.861^{+0.044}_{-0.042}$, $w_a = -0.60^{+0.17}_{-0.19}$

The Unite-only row assumes a constant equation of state. The combined row uses the Chevallier–Polarski–Linder (CPL) form [3, 4]

$$w(a) = w_0 + w_a(1 - a). \quad (1)$$

The numerical reconstruction uses the central values in the combined row. The constant- w result is retained only as a separately labelled comparison. It would be statistically incorrect to combine these values as if they arose from one likelihood. The source paper also reports weak Bayesian evidence for evolving dark energy where frequentist summaries reach roughly 3.1–3.3 σ ; neither statement is evidence specifically for gpt5_6.

We set $\Omega_{r0} = 9 \times 10^{-5}$ and $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ for the demonstration. These are numerical choices rather than additional Unite constraints. The dimensionless background curves do not depend on the numerical value assigned to H_0 .

3 The 16-component field

An ordinary Dirac spinor in 3 + 1 dimensions has four complex components. We define

$$\text{gpt5_6} \equiv \Psi = (\psi_1, \psi_2, \psi_3, \psi_4)^T \in \mathbb{C}^{16}, \quad \Gamma^\mu = I_4 \otimes \gamma^\mu. \quad (2)$$

For metric signature (+ − − −),

$$\{\Gamma^\mu, \Gamma^\nu\} = 2g^{\mu\nu} I_{16}, \quad \bar{\Psi} = \Psi^\dagger \Gamma^0. \quad (3)$$

Thus “16-component” means a reducible four-flavor multiplet, not a new irreducible Lorentz spinor. The flavor-invariant scalar condensate is

$$S = \bar{\Psi}\Psi = \sum_{A=1}^4 \bar{\psi}_A \psi_A. \quad (4)$$

For a homogeneous solution in FLRW, the field equation implies

$$\dot{S} + 3HS = 0, \quad S(a) = S_0 a^{-3}. \quad (5)$$

The condensate therefore fills space and evolves as the universe expands. One positive- Γ^0 representative can be written

$$\Psi(t) = a(t)^{-3/2} e^{-i\Theta(t)} \Psi_0, \quad \dot{\Theta} = U_{,S}. \quad (6)$$

The common phase cancels from S , the vector current, and all background observables used below. The code function `gpt5_6(a)` therefore returns a convenient phase-suppressed representative with four populated positive- Γ^0 components. It is verified to have 16 components, $\bar{\Psi}\Psi = S_0 a^{-3}$, and zero spatial current.

4 Action, field equation, and stress tensor

Use natural units $c = \hbar = 1$, $\kappa = 8\pi G$, and a torsion-free spinor covariant derivative. The action is

$$\mathcal{S} = \int d^4x \sqrt{-g} \left[\frac{R}{2\kappa} + \mathcal{L}_\Psi \right] + \mathcal{S}_m + \mathcal{S}_r, \quad (7)$$

where the Hermitian spinor Lagrangian is

$$\mathcal{L}_\Psi = \frac{i}{2} \left[\bar{\Psi} \Gamma^\mu \nabla_\mu \Psi - (\nabla_\mu \bar{\Psi}) \Gamma^\mu \Psi \right] - U(S). \quad (8)$$

Independent variation of $\bar{\Psi}$ gives

$$i\Gamma^\mu \nabla_\mu \Psi - U_{,S} \Psi = 0, \quad (9)$$

while variation of Ψ gives the adjoint equation. Metric or tetrad variation yields

$$T_{\mu\nu} = \frac{i}{4} \left[\bar{\Psi} \Gamma_\mu \nabla_\nu \Psi + \bar{\Psi} \Gamma_\nu \nabla_\mu \Psi - (\nabla_\mu \bar{\Psi}) \Gamma_\nu \Psi - (\nabla_\nu \bar{\Psi}) \Gamma_\mu \Psi \right] - g_{\mu\nu} \mathcal{L}_\Psi. \quad (10)$$

The two field equations imply $\nabla_\mu T^{\mu\nu} = 0$ on shell. This is the standard nonlinear-spinor construction used in spinor dark-energy models such as Ref. [2], here applied to a four-flavor multiplet.

5 Kinetic density and thermodynamics

Define the first-order Dirac kinetic density

$$K_D = \frac{i}{2} \left[\bar{\Psi} \Gamma^\mu \nabla_\mu \Psi - (\nabla_\mu \bar{\Psi}) \Gamma^\mu \Psi \right]. \quad (11)$$

On the homogeneous equations of motion,

$$K_D = S U_{,S}. \quad (12)$$

The stress tensor becomes the perfect-fluid expression

$$T^\mu{}_\nu = \text{diag}(\rho_\Psi, -p_\Psi, -p_\Psi, -p_\Psi), \quad (13)$$

where

$$\boxed{\rho_\Psi = U(S)}, \quad \boxed{p_\Psi = SU_{,S} - U(S)}, \quad (14)$$

and

$$\boxed{w_\Psi = -1 + \frac{SU_{,S}}{U}}. \quad (15)$$

The on-shell Lagrangian equals the pressure, $\mathcal{L}_\Psi = p_\Psi$.

The word “kinetic” requires care. K_D is a first-order Dirac density; it is not a separately nonnegative contribution like $\dot{\phi}^2/2$ for a canonical scalar. In the reconstructed central history $K_D < 0$ before the $w = -1$ crossing. The trajectory therefore must not be called canonical single-field quintessence.

6 Exact CPL potential reconstruction

Conservation of the spinor component gives

$$\frac{d\rho_\Psi}{d \ln a} = -3[1 + w(a)]\rho_\Psi. \quad (16)$$

Inserting Eq. (1) and integrating produces

$$\rho_\Psi(a) = \rho_{\Psi 0} a^{-3(1+w_0+w_a)} \exp[-3w_a(1-a)]. \quad (17)$$

Using Eq. (5), $a = (S_0/S)^{1/3}$, defines

$$\boxed{U(S) = \rho_{\Psi 0} \left(\frac{S}{S_0}\right)^{1+w_0+w_a} \exp\left[-3w_a\left(1 - \left(\frac{S_0}{S}\right)^{1/3}\right)\right]}. \quad (18)$$

Logarithmic differentiation gives

$$\frac{SU_{,S}}{U} = 1 + w_0 + w_a - w_a \left(\frac{S_0}{S}\right)^{1/3} = 1 + w(a). \quad (19)$$

Equations (14) and (15) therefore reproduce CPL exactly. This demonstrates background equivalence, not perturbative equivalence or uniqueness.

7 Numerical initial-value problem

7.1 Variables and equations

Let $N = \ln a$ and $\tau = H_0(t - t_0)$. The state vector is

$$\mathbf{y}(N) = (\tau, S, \rho_\Psi/\rho_{c0}). \quad (20)$$

Flatness fixes

$$\Omega_{\Psi 0} = 1 - \Omega_{m0} - \Omega_{r0} = 0.69491. \quad (21)$$

The dimensionless Friedmann function and ODEs are

$$E^2(N) \equiv \frac{H^2}{H_0^2} = \Omega_{r0}e^{-4N} + \Omega_{m0}e^{-3N} + \frac{\rho_\Psi}{\rho_{c0}}, \quad (22)$$

$$\frac{d\tau}{dN} = \frac{1}{E}, \quad \frac{dS}{dN} = -3S, \quad \frac{d(\rho_\Psi/\rho_{c0})}{dN} = -3[1 + w(e^N)] \frac{\rho_\Psi}{\rho_{c0}}. \quad (23)$$

At $N = 0$ the initial values are

$$\tau = 0, \quad S = S_0 = 1, \quad \rho_\Psi/\rho_{c0} = 0.69491. \quad (24)$$

The code integrates separately from today to $N = -4$ and to $N = 1$, then joins the output in increasing a . The interval $0.0183 \lesssim a \lesssim 2.718$ contains 1201 output points. SciPy's DOP853 method uses relative tolerance 10^{-11} and absolute tolerance 10^{-13} .

7.2 Discriminating numerical checks

Three analytic identities independently check the integration:

$$S = S_0 a^{-3}, \quad \rho_\Psi = U(S), \quad E^2 = \Omega_{r0}a^{-4} + \Omega_{m0}a^{-3} + U/\rho_{c0}. \quad (25)$$

Table 2 records the maximum relative discrepancies over all 1201 points.

Table 2: Measured numerical consistency errors.

Invariant	Maximum relative error
Bilinear dilution	4.5232×10^{-11}
Density reconstruction	2.2269×10^{-11}
Friedmann closure	1.7988×10^{-12}

These numbers verify the implementation; they are not observational uncertainties.

8 Numerical results

At the present epoch the reconstructed component has

$$\rho_\Psi/\rho_{c0} = 0.69491000, \quad (26)$$

$$K_D/\rho_{c0} = 0.09659249, \quad (27)$$

$$p_\Psi/\rho_{c0} = \mathcal{L}_\Psi/\rho_{c0} = -0.59831751, \quad (28)$$

$$w_\Psi = -0.86100000. \quad (29)$$

The selected equation-of-state values are

$$w(0.1) = -1.401, \quad w(1) = -0.861, \quad w(2) = -0.261. \quad (30)$$

8.1 Phantom crossing and the null-energy boundary

Equation (1) crosses $w = -1$ at

$$a_{w=-1} = 1 + \frac{1+w_0}{w_a} = 0.76833333, \quad z = 0.30151844. \quad (31)$$

Since

$$\rho_\Psi + p_\Psi = SU_{,S} = \rho_\Psi(1 + w), \quad (32)$$

this is also the homogeneous null-energy-condition boundary. The independently interpolated numerical NEC root is $a = 0.76833334$. The central curve has $\rho + p < 0$ before the root and $\rho + p > 0$ after it. The separately evaluated constant- $w = -0.764$ benchmark has no $w = -1$ crossing.

8.2 Acceleration transitions

The deceleration parameter for the complete radiation, matter, and spinor background is

$$q = -\frac{\ddot{a}}{aH^2} = \frac{1}{2} \left(1 + 3 \frac{p_{\text{tot}}}{\rho_{\text{tot}}} \right). \quad (33)$$

Its roots are

Event	a	$z = a^{-1} - 1$
Acceleration begins	0.57194538	0.74841869
Formal future acceleration ends	1.80120338	-0.44481561

The first event is in the observed past. The second follows from extrapolating the empirical CPL law to $a > 1$ and must not be interpreted as an observational forecast.

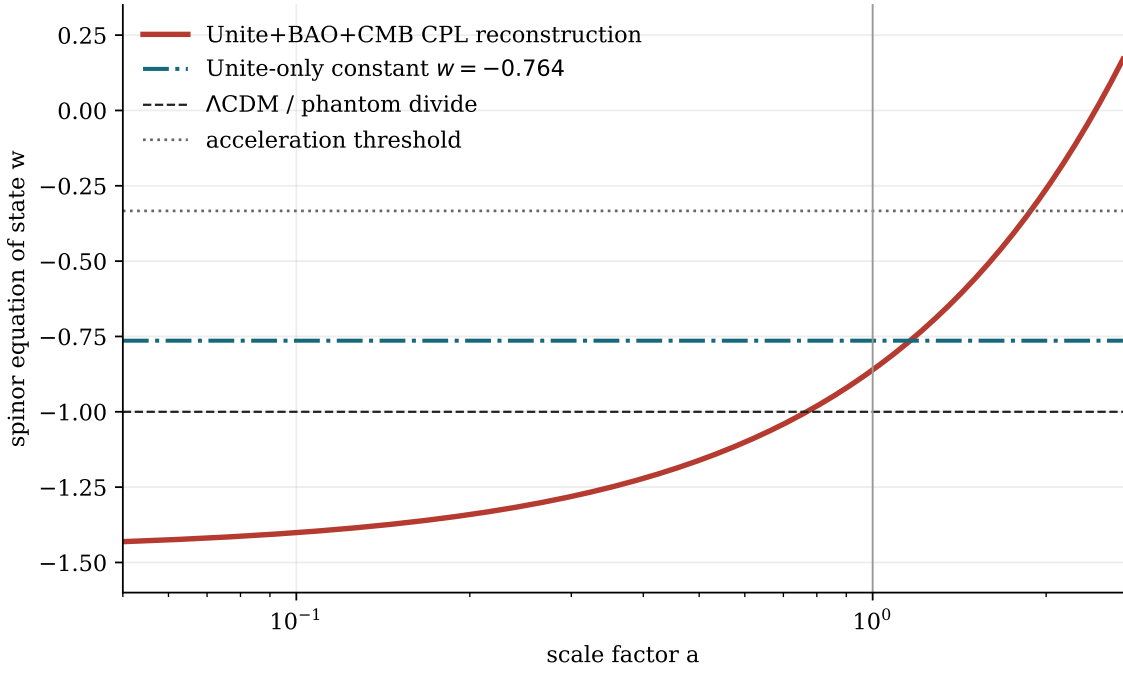


Figure 1: The combined-data CPL central reconstruction, the separate Unite-only constant- w benchmark, and the Λ CDM line. Dataset labels in the legend prevent the two published fits from being conflated.

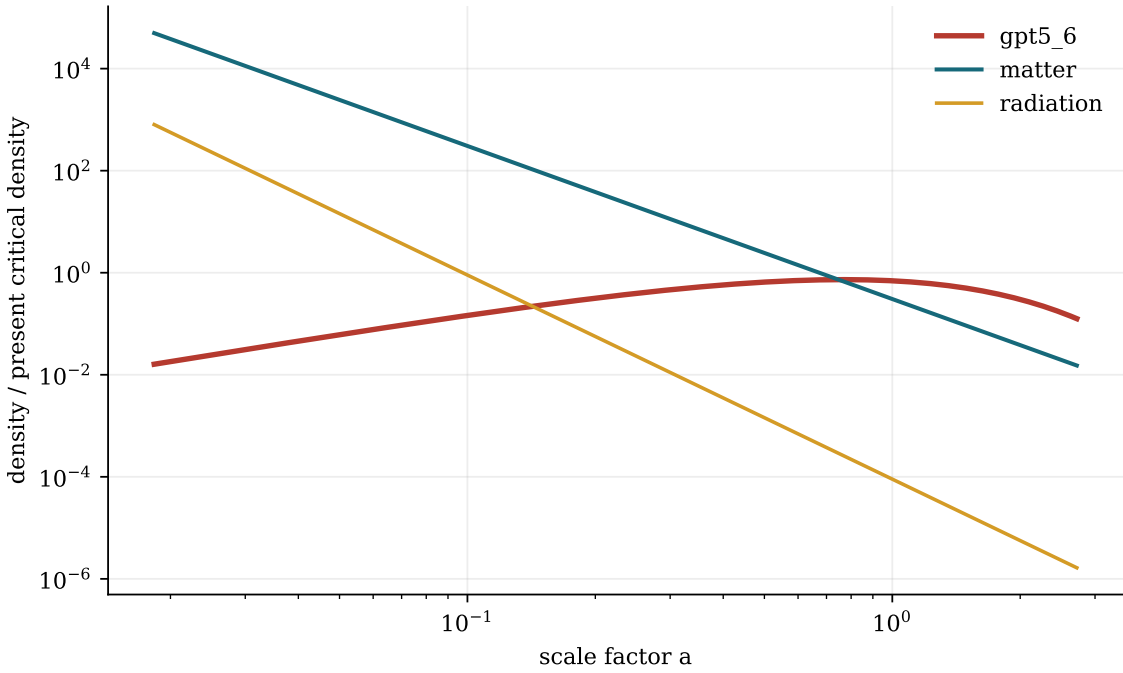


Figure 2: Radiation, matter, and reconstructed spinor densities, all normalized to the present critical density. Matter is a separate fluid in this run.

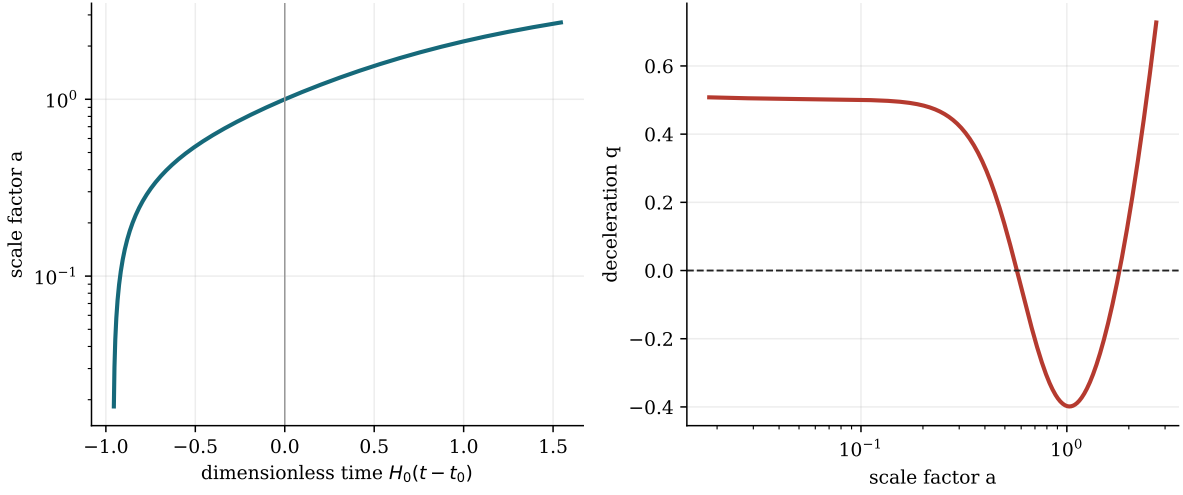


Figure 3: Integrated scale factor and total deceleration parameter. The second $q = 0$ crossing lies in the formal future CPL extrapolation.

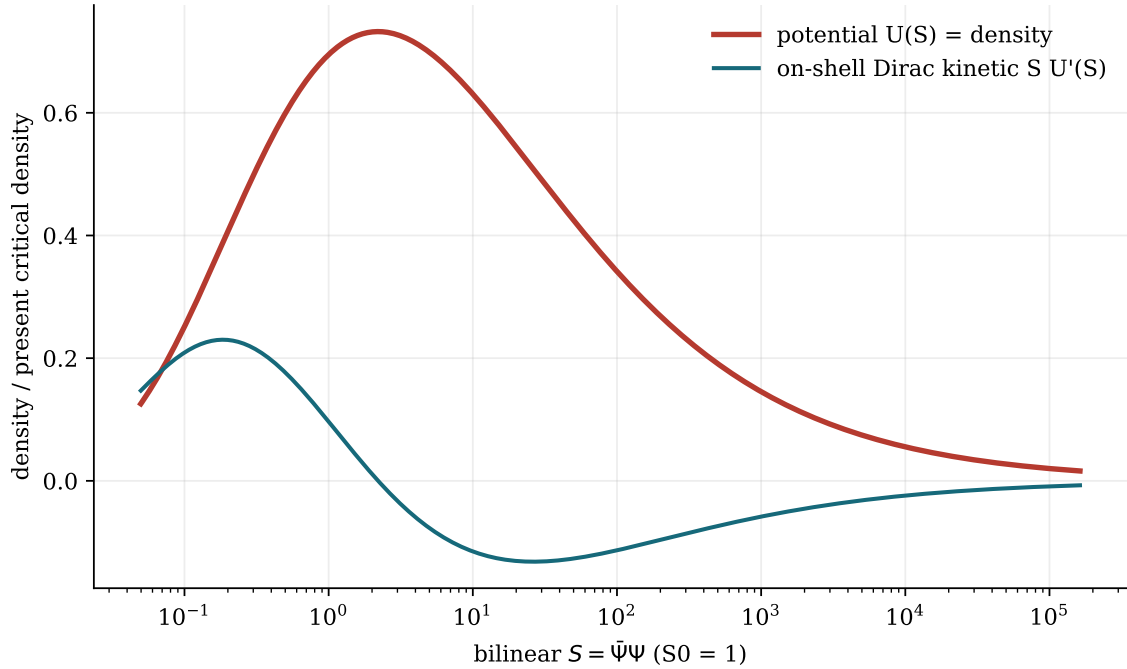


Figure 4: The nonlinear potential $U(S) = \rho_\Psi$ and on-shell first-order Dirac kinetic density $K_D = S U_{,S}$.

9 Dark energy and dark matter

9.1 Dark-energy interpretation

The spinor condensate is homogeneous in the FLRW approximation, has negative pressure today, and makes the present universe accelerate when combined with matter and radiation. It is therefore a consistent effective dark-energy mechanism at background level. Its first-order kinetic structure permits the central reconstructed history to cross $w = -1$, unlike canonical one-field quintessence. This capability does not guarantee viability: a complete fermionic perturbation analysis is needed to determine whether the NEC-violating regime is ghost-free, gradient-stable, causal, and weakly coupled.

9.2 Dark-matter limit

A different potential demonstrates the mathematical connection to dark matter. For

$$U(S) = mS, \tag{34}$$

one obtains

$$p = SU_{,S} - U = 0, \quad w = 0, \quad \rho = mS \propto a^{-3}. \tag{35}$$

The homogeneous component is then dust-like. A potential $U = \Lambda + mS$ can mix constant and dust-like terms. These facts do not make the CPL reconstruction used here dark matter. Equation (22) already includes a separate $\Omega_{m0}a^{-3}$ fluid; identifying the same spinor energy with all matter would double-count it. A unified dark-sector model would require a new fit and successful calculations of clustering, sound speed, CMB anisotropies, lensing, and structure growth.

10 Limitations and open work

- The field is a reducible four-flavor effective multiplet, not a newly discovered irreducible spin representation.
- Only a homogeneous scalar condensate is modeled; isotropy is imposed at the background level.
- The potential is reverse-engineered from a CPL central curve and is not unique.
- No observational likelihood, covariance, nuisance parameters, or uncertainty bands are calculated.
- No scalar, vector, tensor, or fermionic perturbations are evolved.
- Ghost, gradient, causal, radiative, and strong-coupling stability are not established.
- Future CPL behavior is an extrapolation outside the fitted domain.
- $S_0 = 1$ is a background normalization. Particle-physics interpretation requires a dimensionful condensate scale and couplings.

11 Choice of numerical solver

The requested Linux rustSolveIt repository [5] was examined. It vendors a pure-Rust translation of SUNDIALS 7.8.0 and includes a mechanics-oriented notebook language and Jupyter wrapper. It does not expose a drop-in Python ODE interface for the custom state in Eq. (23). Direct use would require a large external workspace, a separate Rust executable, and a Python/Rust data bridge.

This system is smooth, nonstiff, has three variables, and has closed-form invariants. DOP853 reaches relative consistency errors near 10^{-11} , so the extra integration layer offers no numerical benefit for this calculation. A future stiff perturbation system could justify a dedicated CVODE implementation. This is an interface and scope decision, not a claim that SUNDIALS is unsuitable.

12 Reproduction guide

From the repository root on Linux or macOS, create the environment:

```
python3 -m venv .venv
.venv/bin/python -m pip install --upgrade pip
.venv/bin/python -m pip install -r requirements.txt
```

Run the focused tests and regenerate numerical artifacts:

```
.venv/bin/python -m pytest tests/test_gpt5_6_cosmology.py -q
.venv/bin/python scripts/run_cosmology.py
```

Build and execute the notebook, then export HTML:

```
.venv/bin/python scripts/build_cosmology_notebook.py
.venv/bin/python -m jupyter nbconvert --to notebook --execute \
  --inplace notebooks/gpt5_6_cosmology.ipynb \
  --ExecutePreprocessor.timeout=180 \
  --ExecutePreprocessor.record_timing=False
.venv/bin/python -m jupyter nbconvert --to html \
  --output-dir build notebooks/gpt5_6_cosmology.ipynb
```

Build this PDF:

```
bash scripts/build_documentation.sh
```

The expected focused result is 10 passing tests, no notebook error outputs, three invariant errors below 2×10^{-9} , four PNG/PDF figure pairs, and the compiled file docs/gpt5_6_cosmology.pdf. The exact environment, source hashes, and validation commands are recorded in PROVENANCE.md.

References

- [1] R. Camilleri et al., *Supernovae Unite: Combining Pantheon+ and DES-SN5YR*, arXiv:2609.05053v2, especially Sections 7.0.3–7.0.4, <https://arxiv.org/abs/2609.05053>.
- [2] Y.-F. Cai and J. Wang, *Dark Energy Model with Spinor Matter and Its Quintom Scenario*, Class. Quantum Grav. **25** (2008) 165014, arXiv:0806.3890, <https://arxiv.org/abs/0806.3890>.
- [3] M. Chevallier and D. Polarski, *Accelerating Universes with Scaling Dark Matter*, Int. J. Mod. Phys. D **10** (2001) 213–224.

- [4] E. V. Linder, *Exploring the Expansion History of the Universe*, Phys. Rev. D **68** (2003) 083504, arXiv:astro-ph/0208512.
- [5] once-ere, *rustSolveIt on pure-Rust SUNDIALS 7.8.0 for Ubuntu Linux*, https://github.com/once-ere/rustSolveIt_linux_SUNDIALS_7_8_0, accessed 17 September 2026.